

Fundamentals of Mathematics Part I

Module I

Fundamentals of mathematics

- Whole number
- Natural number
- Factor
- Prime number
- Composite number
- Greatest common divisor
- Co prime

Whole number

- A number without a decimal part and it has no negatives.
- All non negative Integers.
- 0, 1, 2, 3, 4, 5, 6, 7

Natural number

- Natural number is an Integer greater than 0.
- It is a member of the set $\{ 1, 2, 3, 4 \dots \}$
- Natural numbers are also known as counting numbers.

Factor

- A factor is a number that divides another number evenly with no remainder.
- Factors are numbers that multiply to get other numbers.
- E.g. 4 is a factor of 8. 4 and 2 are factors of 8. 1 and 8 are factors of 8.
1, 2, 4, 8 are factors of 8.

Prime numbers

- Prime numbers play a very Important role in cryptography.
- They have no other factors except for 1 and the number itself.
- A number that can only be factored as 1 times itself is called “prime”.
- 1 is not a prime number as it does not have distinct factors.
- E.g. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

Composite number

- Numbers that have more than two factors are called as composite numbers.
- Positive Integers that have more than two factors.
- Numbers that are divisible by more than two numbers.
- 4, 6, 8, 9, 10, 12, 14, 15, 16.

Greatest common divisor

- GCD of two or more integers which are not zero is the greatest positive integer that divides each of the Integers.
- GCD of two numbers is the greatest number that divides both the numbers.
- GCD of 12 and 8 is 4.

GCD no.	12	8
2 evenly divides 12, 8	6	4
4 evenly divides 12, 8	3	2

How to find GCD?

- GCD by division:
- First take two of the given numbers, divide the greater by the smaller number and then divide the divisor by the remainder.
- The divisor which does not leave a remainder is the GCD of the two numbers.

$$\begin{aligned} \text{Gcd}(30, 45) &= \\ &= \underline{\underline{15}} \end{aligned} \quad \begin{array}{r} 3 \overline{)45} \begin{array}{l} 15 \\ \hline 30 \\ \hline 15 \end{array} \begin{array}{l} 1 \\ 2 \end{array} \\ 15 \overline{)30} \begin{array}{l} 20 \\ \hline 10 \\ \hline 0 \end{array} \end{array}$$

GCD by division method- Examples

- Find the GCD of 30 and 45.
- Find the GCD of 442 and 546.
- Find the GCD of 442, 546 and 424.

$$\text{GCD}(442, 546) =$$

$$\begin{array}{r} 442 \overline{) 546} (1 \\ \underline{442} \end{array}$$

$$\begin{array}{r} 104 \overline{) 442} (4 \\ \underline{416} \end{array}$$

$$\begin{array}{r} 26 \overline{) 104} (4 \\ \underline{104} \\ \hline 0 \end{array}$$

$$\text{GCD}(442, 546, 424) = 2$$

$$\begin{array}{r} 424 \overline{) 546} (1 \\ \underline{424} \end{array}$$

$$\begin{array}{r} 122 \overline{) 424} (3 \\ \underline{366} \end{array}$$

$$\begin{array}{r} 2 \overline{) 442} (221 \\ \underline{442} \\ \hline 0 \end{array}$$

$$\begin{array}{r} 58 \overline{) 122} (2 \\ \underline{116} \end{array}$$

$$\begin{array}{r} 6 \overline{) 58} (9 \\ \underline{54} \end{array}$$

$$\begin{array}{r} 4 \overline{) 6} (1 \\ \underline{4} \end{array}$$

$$\begin{array}{r} 2 \overline{) 4} (2 \\ \underline{4} \\ \hline 0 \end{array}$$

Co-prime numbers

- Set of numbers that have GCD as 1 are called as Co-prime numbers.
- For instance, 7 and 8 are co-prime numbers.

Properties of Co prime numbers

- 1 is co-prime with every other number.
- Prime numbers are co-prime to each other.
- Any two successive numbers are always co-prime.
- The sum of any two co-prime numbers is always co-prime with their product.
- Examples: 24, 36 are co-primes? **No.** [GCD is not 1]
- 20, 21 are co-primes? **Yes.** [Successive numbers]
- 542, 446 are co-primes? **No.** [GCD is not 1.]